Homework — Bayes factor test suite

Purpose

Practice test-driven development by writing an elaborate test suite for a Bayes-factor object and a minimal working implementation.

Background

Consider binomial data with n trials and k successes, and success probability θ.

Two models differ by prior on θ:

- slab prior: $\theta \sim U(0, 1)$
- spike prior: $\theta \sim U(a, b)$ (where a and b just above and just below the key value 0.5, e.g. $a = 0.4999$ and $b = 0.5001$)

The Bayes factor compares evidence:

- $B = \frac{p(k|\text{spike})}{p(k|\text{slab})}$

Mathematical background

Notation and framing in this section follow Etz, Haaf, Rouder, and Vandekerckhove (2018).

Let the observed data be $x = (k, n)$ where k is successes out of n Bernoulli trials.

For a success-rate parameter θ, the binomial sampling model is

$$P(x \mid \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}.$$

Hypotheses and priors

Define two hypotheses as models on θ:

- $\mathcal{H}_0$ (slab): $\theta \sim U(0, 1)$
- $\mathcal{H}_1$ (spike): $\theta \sim U(a, b)$ (take a and b from the running example in Etz et al., 2018)

So the corresponding prior densities are:

$$p(\theta \mid \mathcal{H}_0) = 1 \text{ on } [0, 1], \qquad p(\theta \mid \mathcal{H}_1) = \frac{1}{c} \text{ on } [a, b].$$

Marginal likelihood and Bayes factor

For model $\mathcal{H}_i$, the model evidence is

$$P(x \mid \mathcal{H}_i) = \int P(x \mid \theta)\, p(\theta \mid \mathcal{H}_i)\, d\theta.$$

Define the Bayes factor favoring $\mathcal{H}_1$ over $\mathcal{H}_0$ as

$$B_{1:0} = \frac{P(x \mid \mathcal{H}_1)}{P(x \mid \mathcal{H}_0)}.$$

Prior odds to posterior odds

Using the notation from the paper:

$$\frac{P(\mathcal{H}_1 \mid x)}{P(\mathcal{H}_0 \mid x)} = \frac{P(\mathcal{H}_1)}{P(\mathcal{H}_0)} \times B_{1:0}.$$

So $B_{1:0}$ is the multiplicative update from prior odds to posterior odds.

Closed-form evidence for this pair

With the two uniforms above:

$$P(x \mid \mathcal{H}_0) = \int_0^1 \binom{n}{k} \theta^k (1-\theta)^{n-k} \, d\theta,$$

$$P(x \mid \mathcal{H}_1) = \int_a^b \frac{1}{c} \binom{n}{k} \theta^k (1-\theta)^{n-k} \, d\theta.$$

Integration in Python To integrate a function in Python, you can use the `scipy.integrate.quad` function. We'll probably talk more about numerical integration later.

```
import scipy.integrate

def integrand(self, x):
    return x * x

result = scipy.integrate.quad(integrand, 0, 1)
print(result)
```

Useful smoke tests for your test suite

- If both models use the same prior, then $B_{1:0} = 1$.
- Evidence terms must be non-negative.
- Invalid inputs ($k > n$, negative counts, invalid θ) should fail predictably.
- Design your own.

Assignment constraints

- ☐ Use Python and `unittest`.
- ☐ Put tests in `tests/test_bayes_factor.py`.
- ☐ Put implementation in `bayes_factor.py`.
- ☐ Provide a **working implementation** of the `BayesFactor` class.
- ☐ The code does **not** need to be elegant, polished, or well-structured.
- ☐ Grading emphasis is on correctness and testing, not software style.

Required Application Programming Interface (API) for the `BayesFactor` class

Implement a class named `BayesFactor` with at least the following methods (you may add more):

- constructor: `BayesFactor(n, k)`
- `likelihood(theta)`
- `evidence_slab()`
- `evidence_spike()`
- `bayes_factor()`

What your tests must cover

1) Input and state validation

☐ Reject invalid `n` and `k` values (type and range checks).
☐ Reject impossible binomial states (e.g., `k > n`).
☐ Ensure object state is consistent after construction.

2) API behavior and return contracts

☐ Each required method exists and is callable.
☐ Methods return the expected data type/shape.

3) Mathematical consistency checks

☐ `likelihood(theta)` behaves sensibly at obvious points.
☐ `bayes_factor()` handles edge cases without crashing.
☐ Tests verify consistency between outputs and your documented mathematical choices.

4) Error behavior

☐ Invalid `theta` values are handled predictably.
☐ Impossible computations raise clear exceptions.
☐ Tests verify exception messages for multiple error cases.

5) TDD quality

☐ Tests are small, focused, and named clearly.
☐ Include at least one fixture or shared setup pattern.
☐ Include at least one intentionally failing test.

Deliverables

Submit via GitHub, using the same repository as before, with the following files:

- `bayes_factor/tests/test_bayes_factor.py`
- `bayes_factor/bayes_factor.py` (working implementation with the required API)
- `bayes_factor/Dockerfile` to create a Docker image that supports your code
- `bayes_factor/README.md` (with not too much detail, just how to run the tests)

Citation

Etz, A., Haaf, J. M., Rouder, J. N., & Vandekerckhove, J. (2018). Bayesian inference and testing any hypothesis you can specify. *Advances in Methods and Practices in Psychological Science, 1*(2), 281-295. https://doi.org/10.1177/2515245918773087